Ali A. Alhejab

Agentic AI | LLMs | Image & Video Generation | Deep Learning | Computer Vision
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PROFESSIONAL EXPERIENCE

Lead AI Engineer — HUMAIN, Riyadh, Saudi Arabia *May 2025 – Present*

- Built an in-house avatar platform, achieving a 92% cost reduction versus third-party solutions.
- Developed and deployed real-time AI avatar agents as interactive virtual presenters and meeting facilitators, with conversational pipelines integrating STT, LLM, and TTS.
- Pretrained and fine-tuned video diffusion models for custom avatar generation, optimizing inference for both CPU and GPU.
- Built an in-house intent-detection model achieving 99% accuracy to route user requests within the conversational pipeline.

AI Engineer — SDAIA, Saudi Arabia *Jun 2024 – May 2025*

Research and development of AI models across computer vision, NLP, and generative modeling.

- Trained DETR and YOLO object-detection models for construction-site surveillance (helmets, vehicles, license plates, and people).
- Built a sign-language-to-text translation system using T5 and AV-HuBERT; contributed to the Sign4All Challenge.
- Fine-tuned LoRA modules for diffusion models (Flux, Hunyuan, Wan 2.1) for image and video generation.
- Fine-tuned LLaMA and T5 for token-by-token image generation/understanding and conversational abilities.
- Optimized inference with TensorRT (vision) and vLLM (LLMs); proficient with TeaCache for generation workflows.
- Built scalable multi-GPU / multi-node training pipelines using HuggingFace Accelerate and DeepSpeed.

Solar Energy Systems Researcher — KAUST, Saudi Arabia *Aug 2022 – May 2024*

Developed techniques to localize and identify faults in PV systems.

- Collaborated with the KAUST Solar Center to build a fast, cost-effective fault-localization algorithm for PV systems.
- Partnered with NEOM to test new solar-glass technologies.

Technical Services Division Trainee — Ithra, Saudi Aramco *Jun 2021 – Aug 2021*

- Member of the quality and maintenance inspection team during a summer internship.

EDUCATION

King Abdullah University of Science and Technology (KAUST) — Thuwal, Saudi Arabia *Jun 2024*

- M.Sc., Electrical and Computer Engineering — Specialization in Solar Energy Systems. GPA: 3.67/4.0.
- Thesis: “A Novel Fault Localization Technique for PV Systems Using a Single-Voltage Sensor.” Advisor: Prof. Shehab Ahmed.

King Fahd University of Petroleum and Minerals (KFUPM) — Dhahran, Saudi Arabia *May 2022*

- B.Sc., Electrical Engineering — Specialization in Internet of Things. GPA: 3.75/4.0.
- Research: “Optimized Synthesis Algorithm for a Two-Phase Switched-Capacitor Converter.” Advisor: Dr. Yaqub Mahnashi.

HONORS & AWARDS

- **1st Place** — Abu Dhabi University 11th Research Competition, 2023. Project: Optimized Synthesis Algorithm for a Two-Phase Switched-Capacitor Converter.
- **1st Place** — KFUPM Senior Design Project EXPO, 2022. Project: Low-Power IoT Data Aggregation System Using a UAV as Data Aggregator.

CERTIFICATES

- **NVIDIA Certified Associate** — Generative AI (Multimodal) & Generative AI (LLM). Riyadh, Saudi Arabia, Dec 2024.
- **Shentech Bootcamp** — Shenzhen, China, Jan 2024. Factory and company visits plus entrepreneurship coursework.
- **PV Design & Installation School** — Thuwal, Saudi Arabia, Apr 2023. 80-hour program on the design, installation, and commissioning of PV systems.

- **Self-Driving Car Engineer Nanodegree** — Online, Jun–Oct 2021. Misk-sponsored, project-based program covering control systems, machine learning, and sensor fusion.

SELECTED PUBLICATIONS

- A. Alhejab et al. (2026). *Saudi Sign Language Translation Using T5*. Springer, Cham.
- A. Alhejab, M. Abbasi, S. Ahmed (2025). *A Single-Voltage-Sensor Bypass-Switch-Based Photovoltaic Fault Localization*. IEEE Journal of Photovoltaics.
- O. Khalifa, A. S. Mohammed, A. Alhejab, et al. (2024). *Energy-Efficient Aerial Data Aggregation for IoT: From Prototyping to Large-Scale Implementation*. IEEE Transactions on Instrumentation and Measurement.
- A. S. Mohammed, O. Khalifa, A. Alhejab, et al. (2023). *Energy-Efficient Aerial Data Aggregation in IoT Networks with WuR*. IEEE International Black Sea Conference on Communications and Networking.
- A. Alhejab, Y. Mahnashi (2024). *A New Synthesis Method for Efficient Reconfigurable Switched-Capacitor Converters*. Authorea Preprints.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB

Software & Tools: HuggingFace, LLaMA-Factory, PyTorch, Git, Jupyter, KiCad, LTspice, LaTeX

Relevant Coursework: Deep Learning for Visual Computing, Mathematics for Machine Learning, Renewables, Microwave Circuits, Computer Networks, IoT, Energy Harvesting

Teaching: Deep Learning and Reinforcement Learning

Languages: Arabic (native), English (advanced)